

Sep 29, 2024 (Due: 08:00 Oct 13, 2025)

1. It can be shown that Gaussian elimination without pivoting is numerically stable for solving strictly diagonally dominant linear systems, in the sense that the growth factor is bounded. Give a concrete upper bound on the growth factor.
2. We have mentioned in the lecture that for complex vectors $x, y \in \mathbb{C}^n$, $\|x\|_2 = \|y\|_2 > 0$ does *not* guarantee that there exists a Householder reflection H such that $y = Hx$. Provide a sufficient and necessary condition for the existence of a Householder reflection H so that $y = Hx$. Prove your claim.

3. Let

$$A = \begin{bmatrix} \alpha_1 & \rho_2 & \rho_3 & \cdots & \cdots & \rho_n \\ \beta_2 & \alpha_2 & 0 & \cdots & \cdots & 0 \\ \beta_3 & 0 & \alpha_3 & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & & \ddots & \alpha_{n-1} & 0 \\ \beta_n & 0 & \cdots & \cdots & 0 & \alpha_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Design an efficient algorithm to compute the QR factorization of A .

4. Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. Show that Q can be factorized as the product of finitely many Householder reflections, and if, in addition, $\det(Q) = 1$, Q can be factorized as the product of finitely many Givens rotations.

5. Solve the following linear systems, using Gaussian elimination with and without pivoting:

$$\begin{bmatrix} 8 & 1 & & & & \\ 6 & 8 & 1 & & & \\ & 6 & 8 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 6 & 8 & 1 \\ & & & & 6 & 8 & 1 \\ & & & & & 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{98} \\ x_{99} \\ x_{100} \end{bmatrix} = \begin{bmatrix} 9 \\ 15 \\ 15 \\ \vdots \\ 15 \\ 15 \\ 14 \end{bmatrix}$$

and

$$\begin{bmatrix} 6 & 1 & & & & \\ 8 & 6 & 1 & & & \\ & 8 & 6 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 8 & 6 & 1 \\ & & & & 8 & 6 & 1 \\ & & & & & 8 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{98} \\ x_{99} \\ x_{100} \end{bmatrix} = \begin{bmatrix} 7 \\ 15 \\ 15 \\ \vdots \\ 15 \\ 15 \\ 14 \end{bmatrix}.$$

Compare the computed solutions with the exact ones. How can you say about the accuracy?

6. (H) Implement the cyclic reduction algorithm for solving tridiagonal linear systems. Test your implementation with the examples in Exercise 5.

(optional) How to solve banded linear systems using cyclic reduction?

7. (optional) It can be shown that Gaussian elimination with partial pivoting is numerically stable for solving nonsingular tridiagonal linear systems, in the sense that the growth factor is bounded. Give a concrete upper bound on the growth factor.

8. (optional) Let $w_1, w_2, \dots, w_k \in \mathbb{C}^n$ be unit vectors. Try to find a matrix $T \in \mathbb{C}^{k \times k}$ such that

$$(I - 2w_1w_1^*)(I - 2w_2w_2^*) \cdots (I - 2w_kw_k^*) = I - [w_1, w_2, \dots, w_k]T[w_1, w_2, \dots, w_k]^*.$$